



2IA Practice Problems II

Permutations



Practice Problems Chapter 2

1. Consider the following permutations in S_5 :

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 5 & 3 & 4 \end{pmatrix}, \quad \beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 1 & 2 \end{pmatrix}, \quad \gamma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 2 & 1 & 3 \end{pmatrix}$$

Compute:

- (a) α^{-1}
- (b) $\alpha\beta^2$
- (c) $\gamma\beta^{-1}$
- (d) $\alpha^{-1}\gamma\beta^{-1}$

Solution: Proceeding like in the notes we obtain:

$$(a) \quad \alpha^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 4 & 5 & 3 \end{pmatrix}$$

$$(b) \quad \beta^2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 3 & 5 & 4 \end{pmatrix}, \quad \alpha\beta^2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 5 & 4 & 3 \end{pmatrix}$$

$$(c) \quad \beta^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 3 & 2 & 1 \end{pmatrix}, \quad \gamma\beta^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 2 & 5 & 4 \end{pmatrix}$$

$$(d) \quad \alpha^{-1}\gamma\beta^{-1} = \alpha^{-1}(\gamma\beta^{-1}) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 1 & 4 & 3 \end{pmatrix}$$

2. Let $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$ be permutations in S_4 . In each case solve the given equation for χ in S_4 .

- (a) $\chi\tau = \sigma$
- (b) $\sigma^{-1}\chi = \tau$
- (c) $\chi\tau\sigma = \varepsilon$
- (d) $\tau\chi\sigma^{-1} = \sigma$

Solution: Let us begin by computing σ^{-1} and τ^{-1} :

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix}, \quad \tau^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$$

$$(a) \quad \chi = \sigma\tau^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 2 & 4 \end{pmatrix}$$

$$(b) \quad \chi = \sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 2 & 4 \end{pmatrix}$$

$$(c) \quad \chi\tau = \sigma^{-1} \text{ and so } \chi = \sigma^{-1}\tau^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix}$$

$$(d) \quad \chi = \tau^{-1}\sigma^2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}, \text{ since } \sigma^2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$$

3. Write the following permutations in to matrix-type notation.

- (a) $\sigma = (1 \ 8 \ 7 \ 4)(3 \ 6 \ 7 \ 5 \ 9)$
- (b) $\tau = (1 \ 3 \ 5 \ 7)(4 \ 1 \ 9)$
- (c) $\gamma = (7 \ 1 \ 8 \ 9 \ 2 \ 5 \ 6)(3 \ 6)(4 \ 1 \ 9)$

Solution: (a) $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 8 & 2 & 6 & 1 & 9 & 4 & 5 & 7 & 3 \end{pmatrix}$

$$(b) \quad \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 9 & 2 & 5 & 3 & 7 & 6 & 1 & 8 & 4 \end{pmatrix}$$

$$(c) \quad \gamma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 5 & 7 & 8 & 6 & 3 & 1 & 9 & 4 \end{pmatrix}$$

4. Factor each of the following permutations into disjoint cycles, and into transpositions. In each case, find the parity.

(a) $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 3 & 8 & 9 & 5 & 2 & 1 & 6 & 4 & 7 \end{pmatrix}$

(b) $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 6 & 4 & 8 & 9 & 3 & 1 & 7 & 5 & 2 \end{pmatrix}$

(c) $\gamma = (1 \ 2 \ 3 \ 4 \ 5)(6 \ 7)(1 \ 3 \ 5 \ 7)(1 \ 6 \ 3)$

Solution: (a)

$$\begin{aligned} \sigma &= (1 \ 3 \ 9 \ 7 \ 6)(2 \ 8 \ 4 \ 5) \\ &= (1 \ 3)(3 \ 9)(9 \ 7)(7 \ 6)(2 \ 8)(8 \ 4)(4 \ 5) \\ &= (1 \ 6)(1 \ 7)(1 \ 9)(1 \ 3)(2 \ 5)(2 \ 4)(2 \ 8) \end{aligned}$$

(b)

$$\begin{aligned} \tau &= (1 \ 6)(2 \ 4 \ 9)(3 \ 8 \ 5) \\ &= (1 \ 6)(2 \ 4)(4 \ 9)(3 \ 8)(8 \ 5) \\ &= (1 \ 6)(2 \ 9)(2 \ 4)(3 \ 5)(3 \ 8) \end{aligned}$$

(c) Observe that γ is a product of cycles but they are not disjoint. Thus, in order to find the factorisation of γ into disjoint cycles, we need to write it first in two-row matrix notation.

$$\begin{aligned} \gamma &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 3 & 4 & 5 & 6 & 1 & 2 \end{pmatrix} = (1 \ 7 \ 2 \ 3 \ 4 \ 5 \ 6) \\ &= (1 \ 7)(7 \ 2)(2 \ 3)(3 \ 4)(4 \ 5)(5 \ 6) \\ &= (1 \ 6)(1 \ 5)(1 \ 4)(1 \ 3)(1 \ 2)(1 \ 7) \end{aligned}$$

5. (Test in 2021) Consider the permutation $\sigma \in S_9$, given below

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 5 & 1 & 6 & 8 & 7 & 3 & 9 & 4 & 2 \end{pmatrix}.$$

- (a) Find the decomposition into disjoint cycles of σ .
- (b) What is the length of the longest cycle?
- (c) Swap any two element in the second row of the matrix notation of σ to obtain a permutation $\beta \in S_9$ such that the decomposition into disjoint cycles of β has a cycle of longer length.

Solution: (a) $\sigma = (1\ 5\ 7\ 9\ 2)(3\ 6)(4\ 8)$.

(b) From (a) we can see that σ is a product of two transpositions (that is, 2-cycles) and a 5-cycle. So, the length of the longest cycle is 5.

(c) We have been asked to swap any two elements of the second row of σ in such a way that the decomposition into disjoint cycles of the resulting permutation β has a cycle of length > 5 . To do so, we should change the value of $\sigma(2)$ because $\sigma(2) = 1$ and this takes us to the beginning of that 5-cycle. For example, swapping 1 and 6 in the second row of σ should work. In fact, the resulting permutation is:

$$\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 5 & 6 & 1 & 8 & 7 & 3 & 9 & 4 & 2 \end{pmatrix}.$$

The cycle decomposition of β is as follows:

$$\beta = (1\ 5\ 7\ 9\ 2\ 6\ 3)(4\ 8).$$

It contains a cycle of longer length, namely, 7.

6. **In each case determine whether the statement is true or false.** If true, provide an argument supporting your answer; if false, provide a counterexample.
- (a) The product of transpositions $(1\ 2)(1\ 3)\dots(1\ r)$ is indeed r -cycle.
 - (b) If α and β are disjoint permutations in S_n such that $\alpha\beta = \varepsilon$, then $\beta = \alpha^{-1}$.
 - (c) If $\alpha, \beta, \gamma \in S_n$ are such that $\alpha\beta = \gamma\alpha$, then $\beta = \gamma$.
 - (d) The product of any two cycles commute.
 - (e) If σ and τ are distinct transpositions in S_n , then their product $\sigma\tau$ has always order 2.
 - (f) If σ and τ are disjoint transpositions in S_n , then their product $\sigma\tau$ has always order 2.
 - (g) If α and β are not disjoint, then $\alpha\beta \neq \beta\alpha$.

Solution:

(a) **True:** We proved in the workshop sessions that any r -cycle $(k_1\ k_2\ \dots\ k_r)$ can be expressed as the product of $r - 1$ in two different ways. Here we will use the following one:

$$(k_1\ k_2\ \dots\ k_r) = (k_1\ k_r)(k_1\ k_{r-1})\dots(k_1\ k_2) \quad (1)$$

Now, notice that 1 appears in all the factor transpositions, so one could think about making $k_1 = 1$ and then write $2, 3, \dots, r - 1, r$ in reversed order to obtain the r -cycle: $(1\ r\ r - 1\ \dots\ 3\ 2)$. Using (1) we obtain that

$$(1\ r\ r - 1\ \dots\ 3\ 2) = (1\ 2)(1\ 3)\dots(1\ r - 1)(1\ r),$$

proving the statement.

(b) **True:** Keeping in mind that disjoint permutations commute, from $\alpha\beta = \varepsilon$ we have that $\beta\alpha = \varepsilon$, which shows that $\beta = \alpha^{-1}$.

(c) **False:** In S_4 , take $\alpha = (1\ 2\ 3\ 4)$, $\beta = (1\ 2)$ and $\gamma = (2\ 3)$. Then we have that

$$\alpha\beta = (1\ 3\ 4) = \gamma\alpha,$$

and clearly $\beta \neq \gamma$.

(d) **False:** For instance, take $\alpha = (1\ 3)$ and $\beta = (2\ 3)$ in S_3 . We have that

$$\alpha\beta = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$\beta\alpha = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

Thus α and β do not commute.

(e) **False:** Take for example $\sigma = (1 \ 2)$ and $\tau = (2 \ 3)$. Then you can check that $\sigma\tau = (1 \ 2 \ 3)$, which is a 3-cycle and has order 3.

(f) **True:** First of all notice that $\sigma^2 = \tau^2 = \varepsilon$. On other hand, $\sigma\tau = \tau\sigma$, since we are assuming that σ and τ are disjoint. From here we obtain that

$$(\sigma\tau)^2 = \sigma^2\tau^2 = \varepsilon,$$

which shows that $\sigma\tau$ has order 2.

(f) **False:** We proved in class that disjoint permutations commute. But there are lots of permutations that are not disjoint and commute. I am writing this question here because some of you made that mistake in Test 1. We can obtain a counterexample very easily by taking any α in S_n and $\beta = \alpha^{-1}$. It is obvious that α and β are not disjoint and $\alpha\beta = \varepsilon = \beta\alpha$.

7. (Test 2, 2021) Let $\sigma \in S_n$ be a permutation and $(i_1 i_2 \dots i_k)$ a k -cycle in S_n . Prove that

$$\sigma(i_1 i_2 \dots i_k)\sigma^{-1} = (\sigma(i_1) \sigma(i_2) \dots \sigma(i_k)).$$

Solution: Let $\tau = (i_1 i_2 \dots i_k)$ and $\gamma = (\sigma(i_1) \sigma(i_2) \dots \sigma(i_k))$. To prove that $\sigma\tau\sigma^{-1} = \gamma$, we will show that

$$(\sigma\tau\sigma^{-1})(t) = \gamma(t),$$

for all $t \in X_n$. We distinguish a few cases:

- $t = \sigma(i_r)$ for $r \in \{1, \dots, k-1\}$.

In this case, we have that

$$\begin{aligned} (\sigma\tau\sigma^{-1})(t) &= (\sigma\tau\sigma^{-1})(\sigma(i_r)) = (\sigma\tau)(i_r) = \sigma(\tau(i_r)) = \sigma(i_{r+1}), \\ \gamma(t) &= \gamma(\sigma(i_r)) = \sigma(i_{r+1}). \end{aligned}$$

- $t = \sigma(i_k)$

In this second case, we have that

$$\begin{aligned} (\sigma\tau\sigma^{-1})(t) &= (\sigma\tau\sigma^{-1})(\sigma(i_k)) = (\sigma\tau)(i_k) = \sigma(\tau(i_k)) = \sigma(i_1), \\ \gamma(t) &= \gamma(\sigma(i_k)) = \sigma(i_1). \end{aligned}$$

- $t \neq \{\sigma(i_1), \dots, \sigma(i_k)\}$.

Then $\sigma^{-1}(t) \neq \{i_1, \dots, i_k\}$, and so $\tau(\sigma^{-1}(t)) = \sigma^{-1}(t)$. Thus:

$$\begin{aligned} (\sigma\tau\sigma^{-1})(t) &= \sigma(\tau(\sigma^{-1}(t))) = \sigma(\sigma^{-1}(t)) = t, \\ \gamma(t) &= t. \end{aligned}$$